

DIFFERENCES BETWEEN DECIDUOUS AND PERMANENT POSTERIOR DENTITION

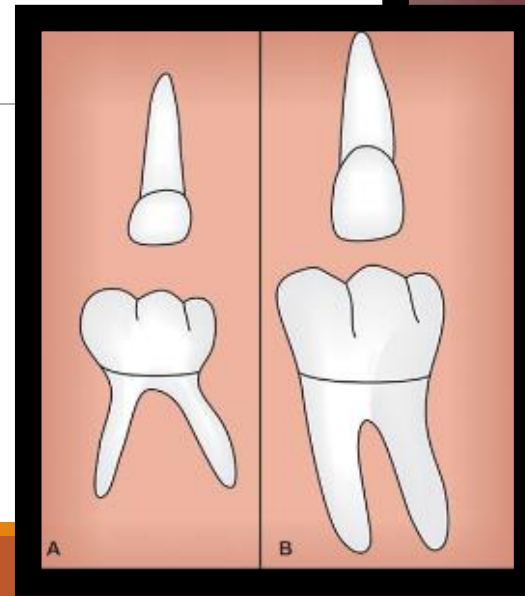
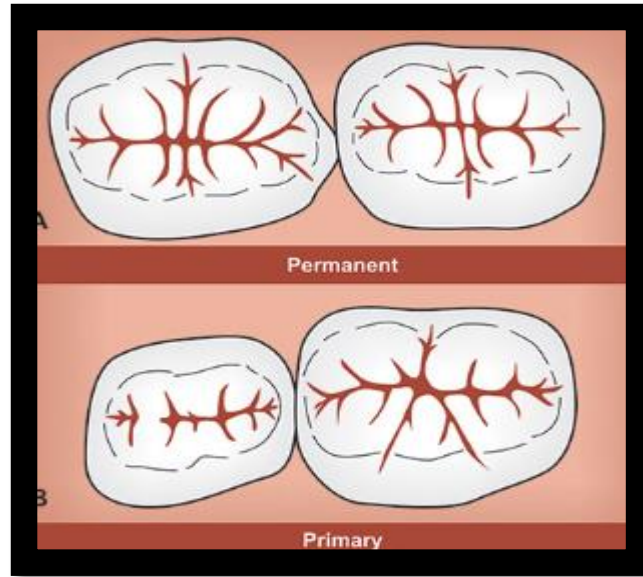
DR. MISBAH ASHRAF MOTEN

LECTURER AND MDS TRAINEE

(ORAL BIOLOGY) DIKIOHS, DUHS

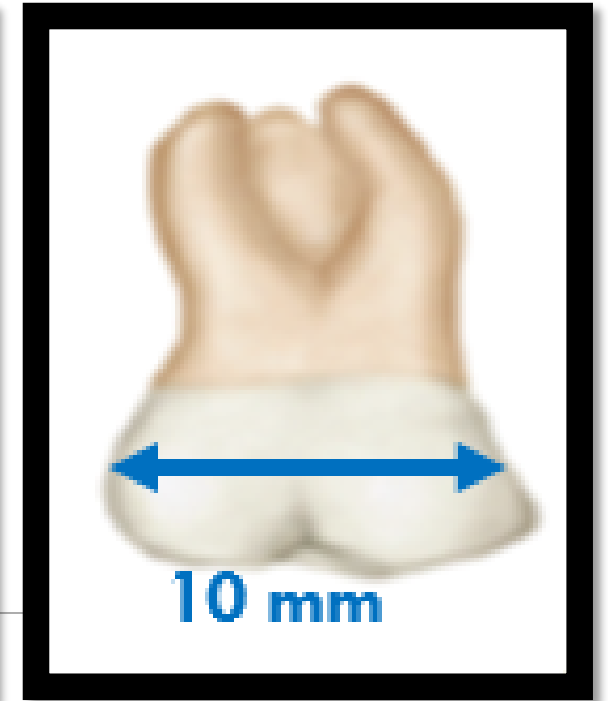
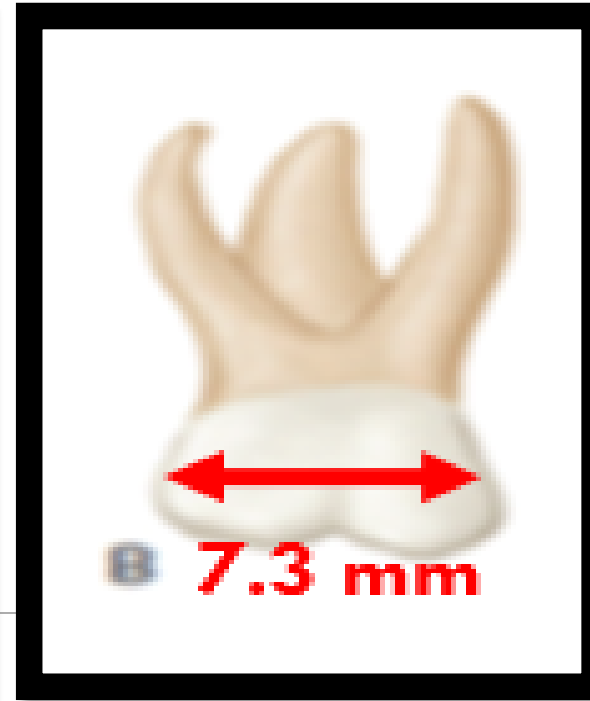
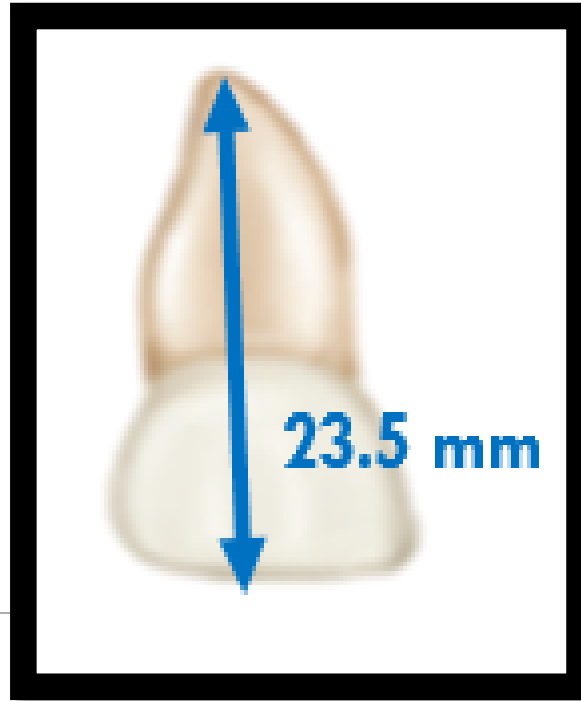


GENERAL DIFFERENCES BETWEEN DECIDUOUS AND PERMANENT DENTITION



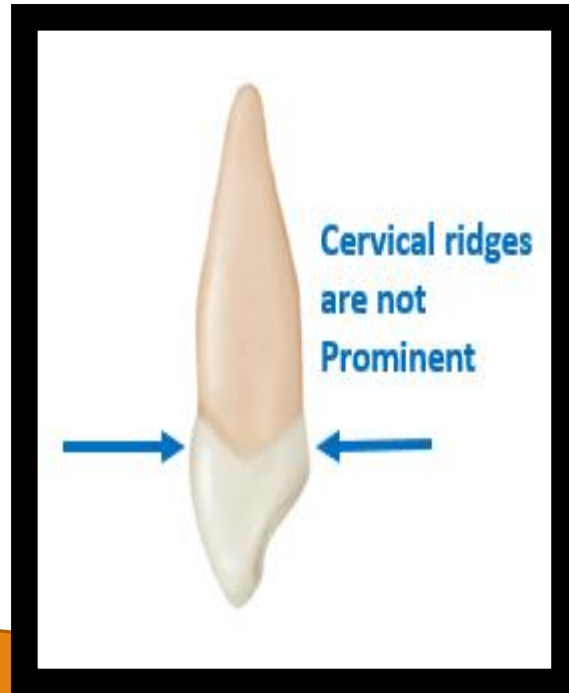
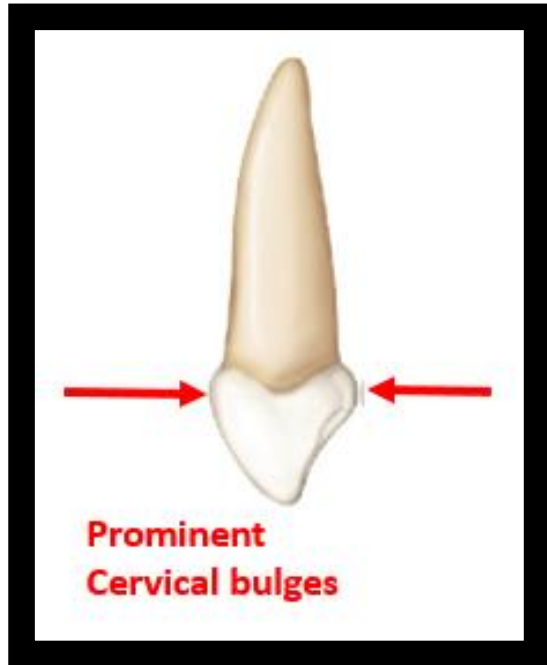
IDENTIFY THE SHOWN
DIFFERENCES IN THE
IMAGES

SIZE & CROWN DIMENSIONS



Primary teeth in comparison to permanent counterparts are smaller in overall size & crown dimensions.

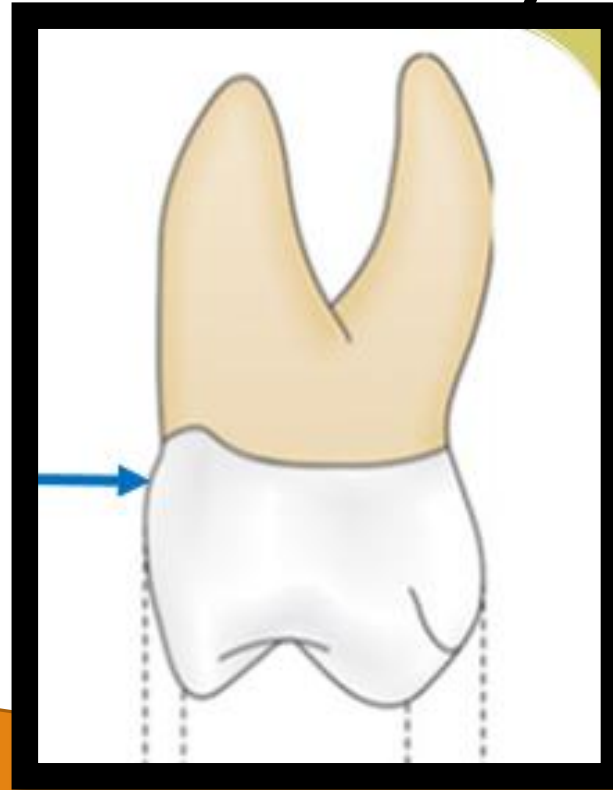
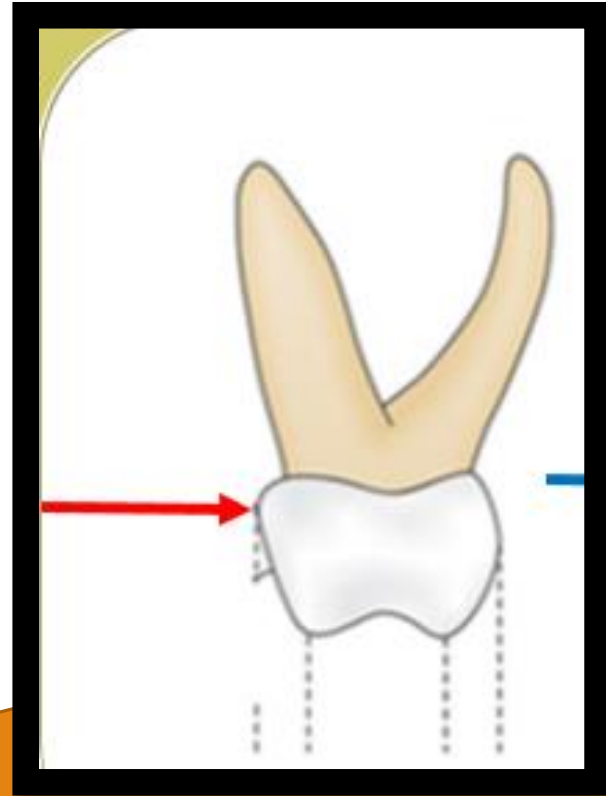
CERVICAL RIDGES (ANTERIORS)



Carefully
considered
during
operative
procedures

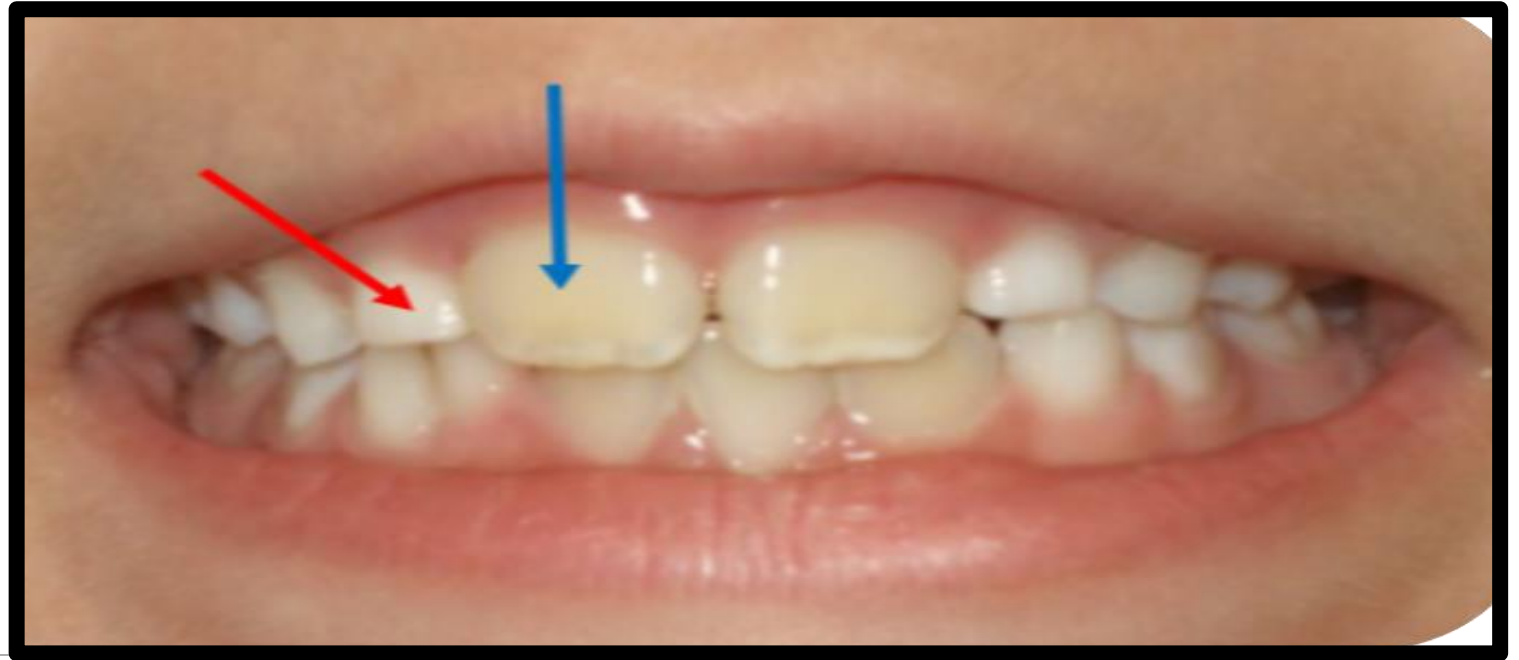
Primary teeth as compared to permanent counterparts have markedly more prominent cervical ridges, narrower necks.

CERVICAL RIDGES (POSTERIORS)



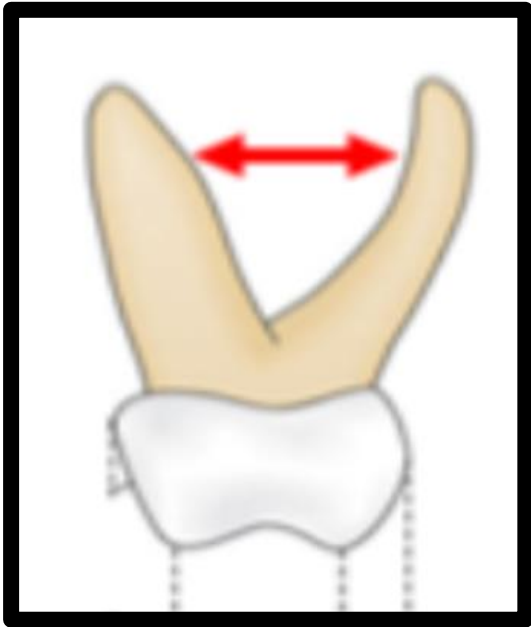
Buccal cervical ridges on primary molars are much more pronounced as compared to permanent counterparts, especially on maxillary & mandibular 1st molars

COLOR

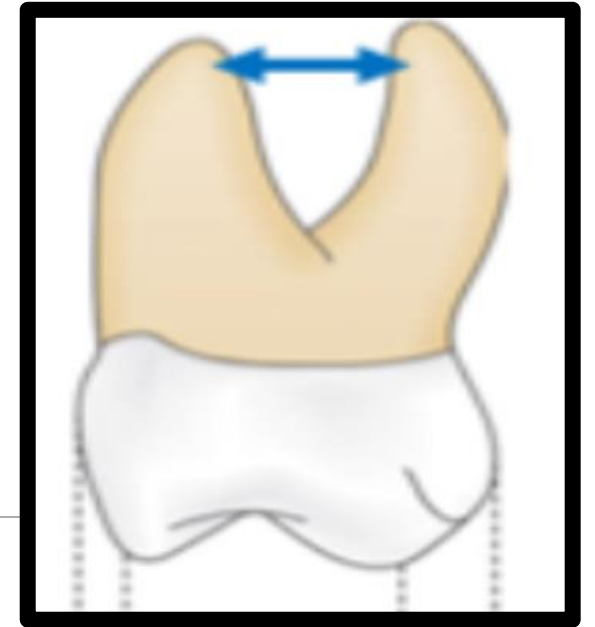


Primary teeth are lighter in color clinically with less pigmentation & the appearance of primary teeth is milky white WHEREAS, Permanent teeth are darker in color appears yellowish white to greyish white

ROOT FLARE

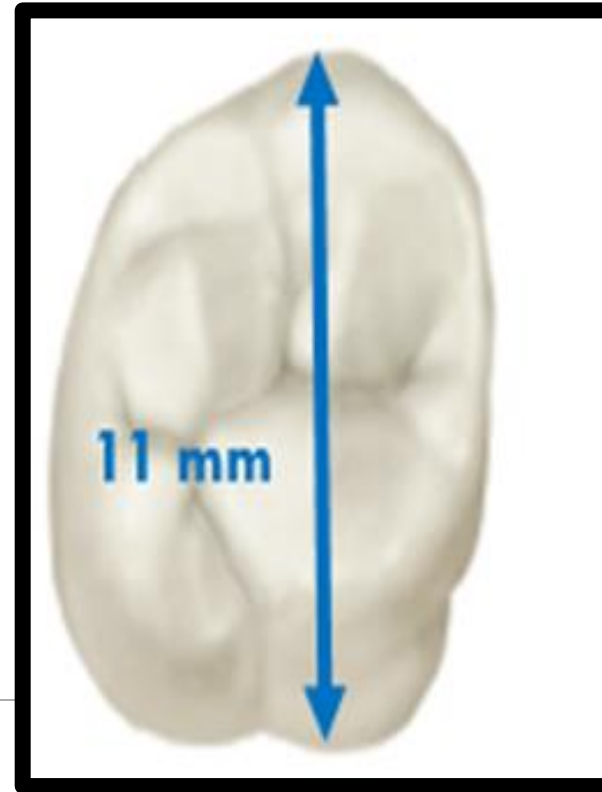
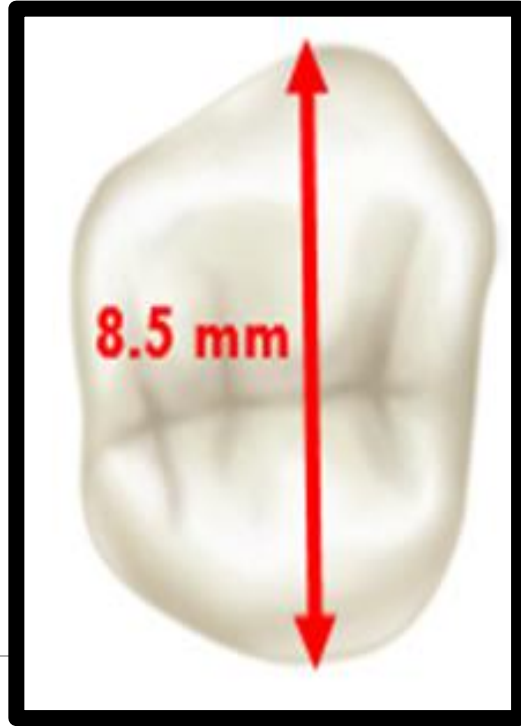


The flare allows more room between the roots for the development of permanent tooth crowns.



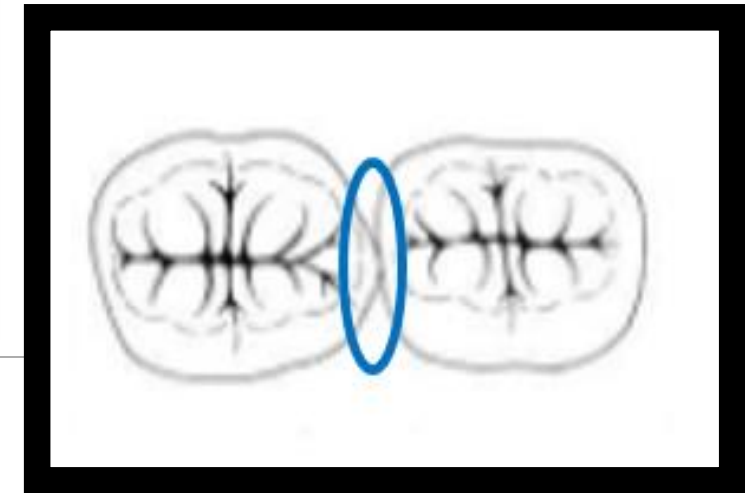
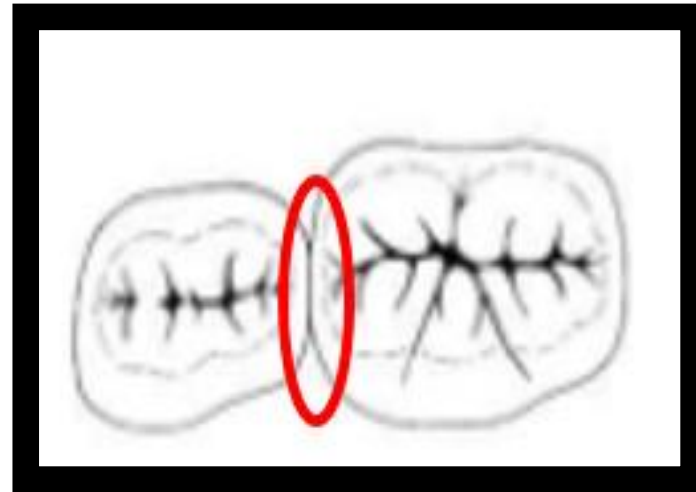
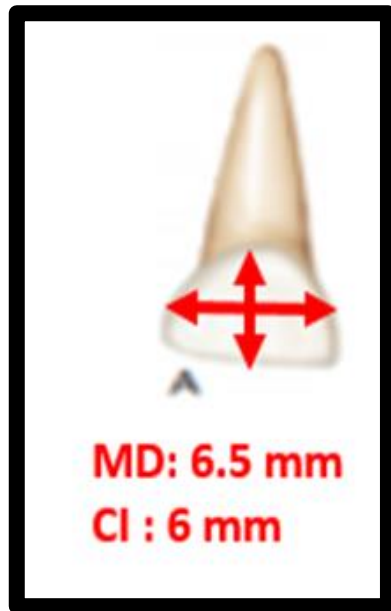
Primary teeth in comparison to permanent counterparts have widely flared and long roots which extend beyond projected outlines of the crowns.

BL DIMENSIONS



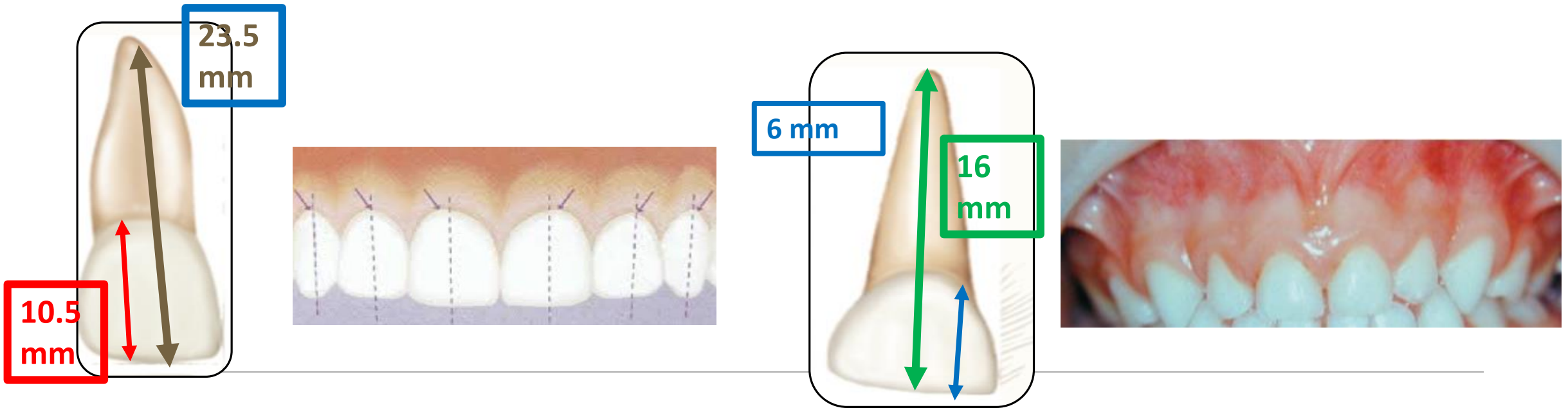
BL diameter of primary molar teeth is less than that of permanent teeth.

CROWN DIMENSIONS (ANTERIORS) & CONTACT AREAS



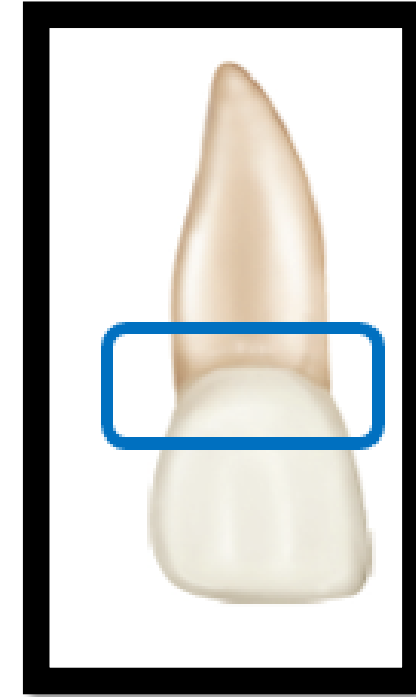
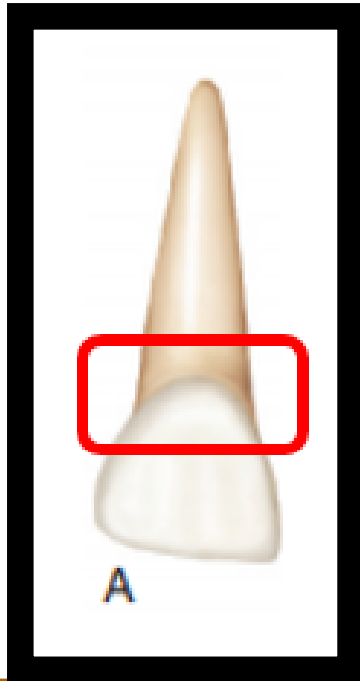
Crowns of primary anterior teeth are wider MD as compared to crown length and has wide contact areas whereas crown length > MD width in case of permanent anteriors and they have pointed contact areas.

CROWN DIMENSIONS (ANTERIORS) & CONTACT AREAS



- primary teeth have ↓ crown height/tooth length ratio as compared to permanent teeth
- Contact areas are broader flatter and more cervically placed in primary anteriors as compared to permanent.

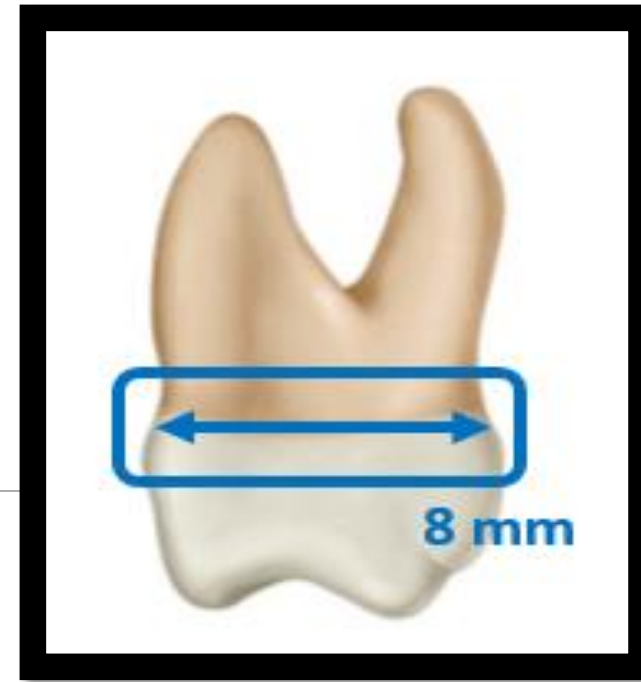
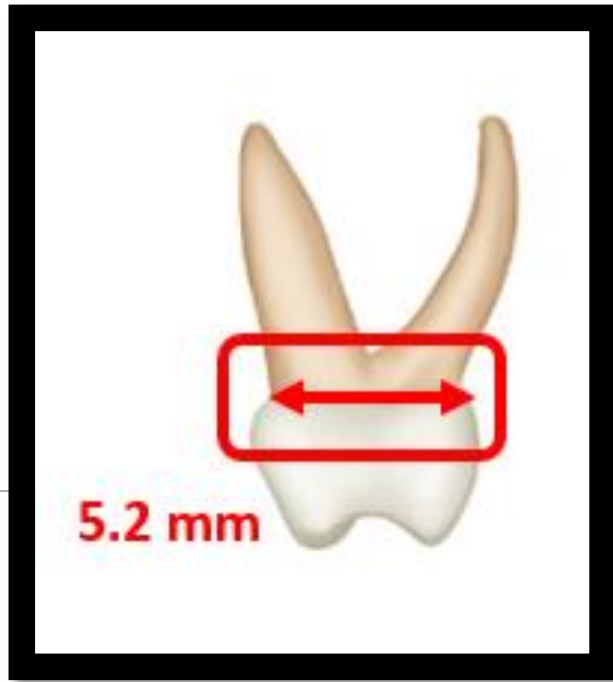
ROOTS AND CERVICAL 3RD (ANTERIORS)



The roots of primary anterior teeth are narrower & longer comparatively.

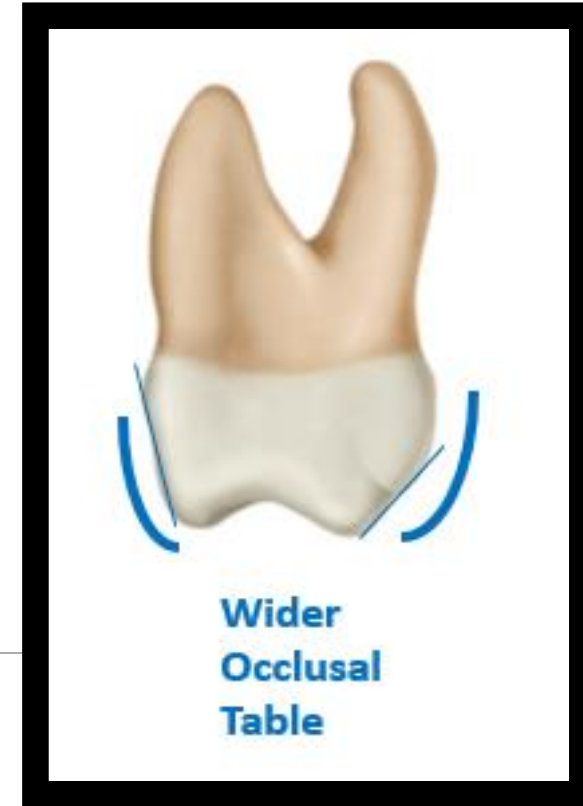
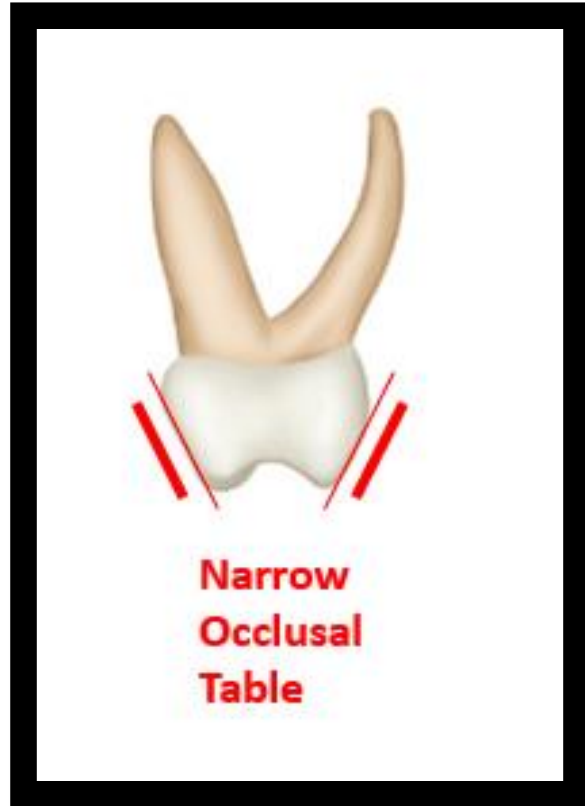
Narrow roots with wide crowns present an arrangement at the cervical 3rd of crown & root that differs markedly from that of the permanent anterior teeth.

CROWNS AND ROOTS OF CERVICAL PORTIONS (POSTERIORS)



MD slender cervical portions for crowns and roots of primary molars as compared to permanent counterparts

B & L SURFACES OUTLINE AND OCCLUSAL TABLE (POSTERIOR)



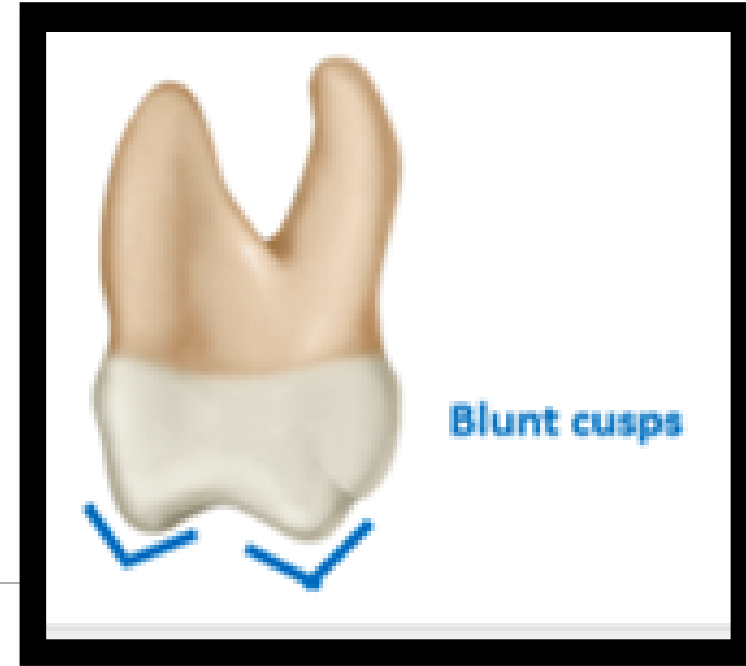
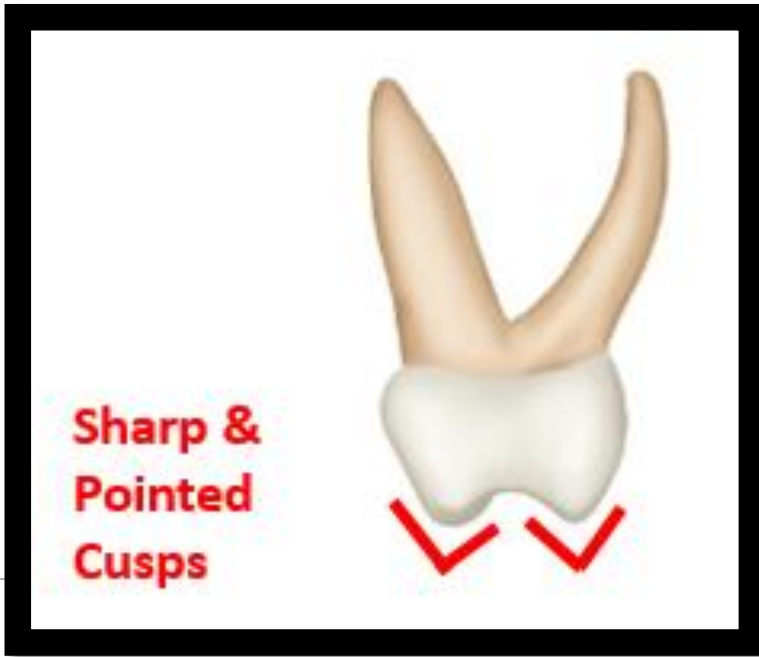
B & L surfaces of primary molars are flatter above cervical curvatures than those of permanent molars, this narrows occlusal surfaces.

PRESENCE OF MAMELONS AND GROOVES



No developmental grooves or mamelons on primary incisors whereas developmental grooves and mamelons (newly erupted incisors) are features of permanent counterparts.

SHARPNESS OF CUSPS



Bulbous crowns with pointed cusps for primary dentition as compared to permanent dentition whose cusps are blunter.

COMPARATIVE EVALUATION OF PULP CHAMBERS AND CANALS OF PRIMARY & PERMANENT TEETH

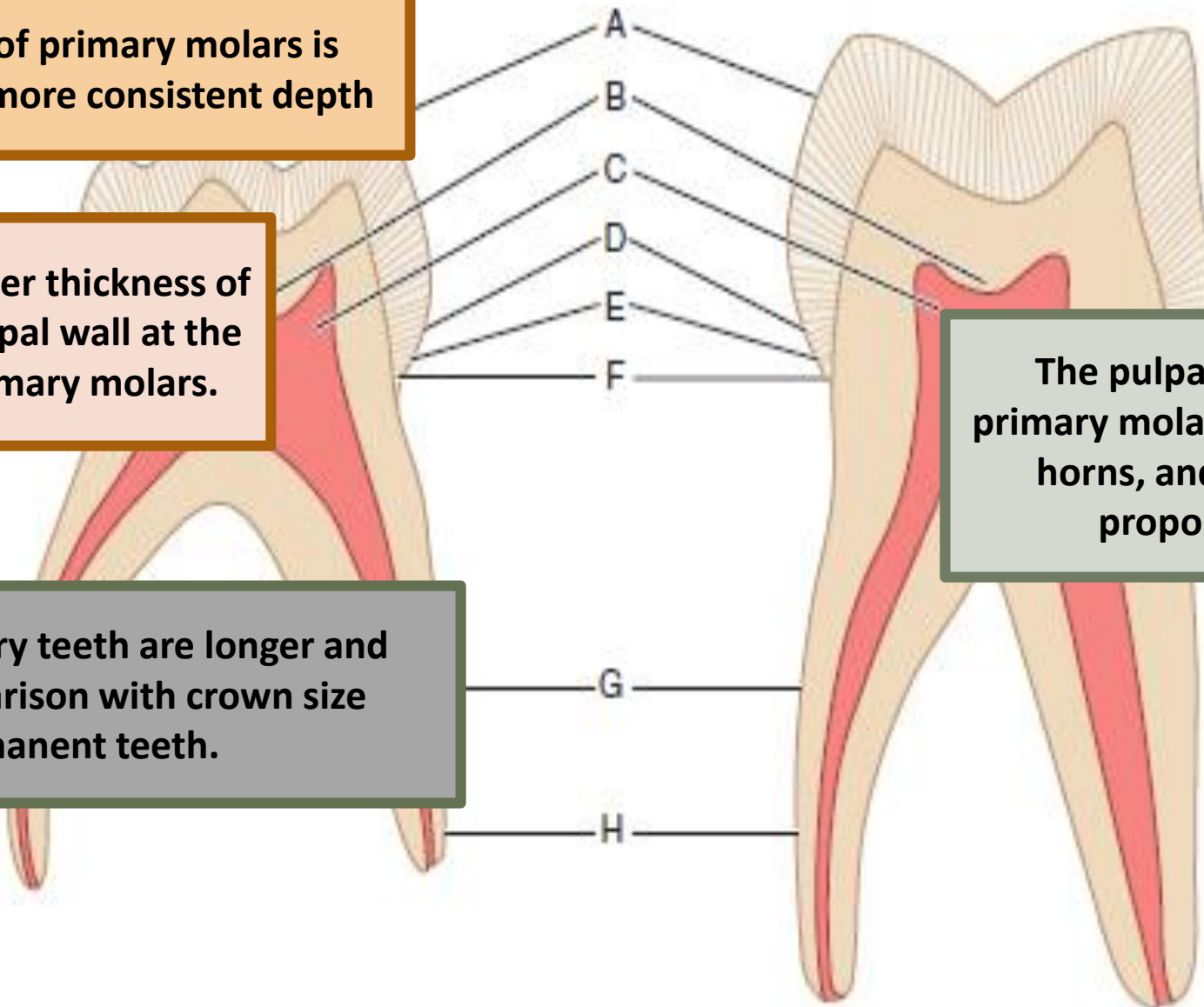
IDENTIFY THE MARKED DIFFERENCES



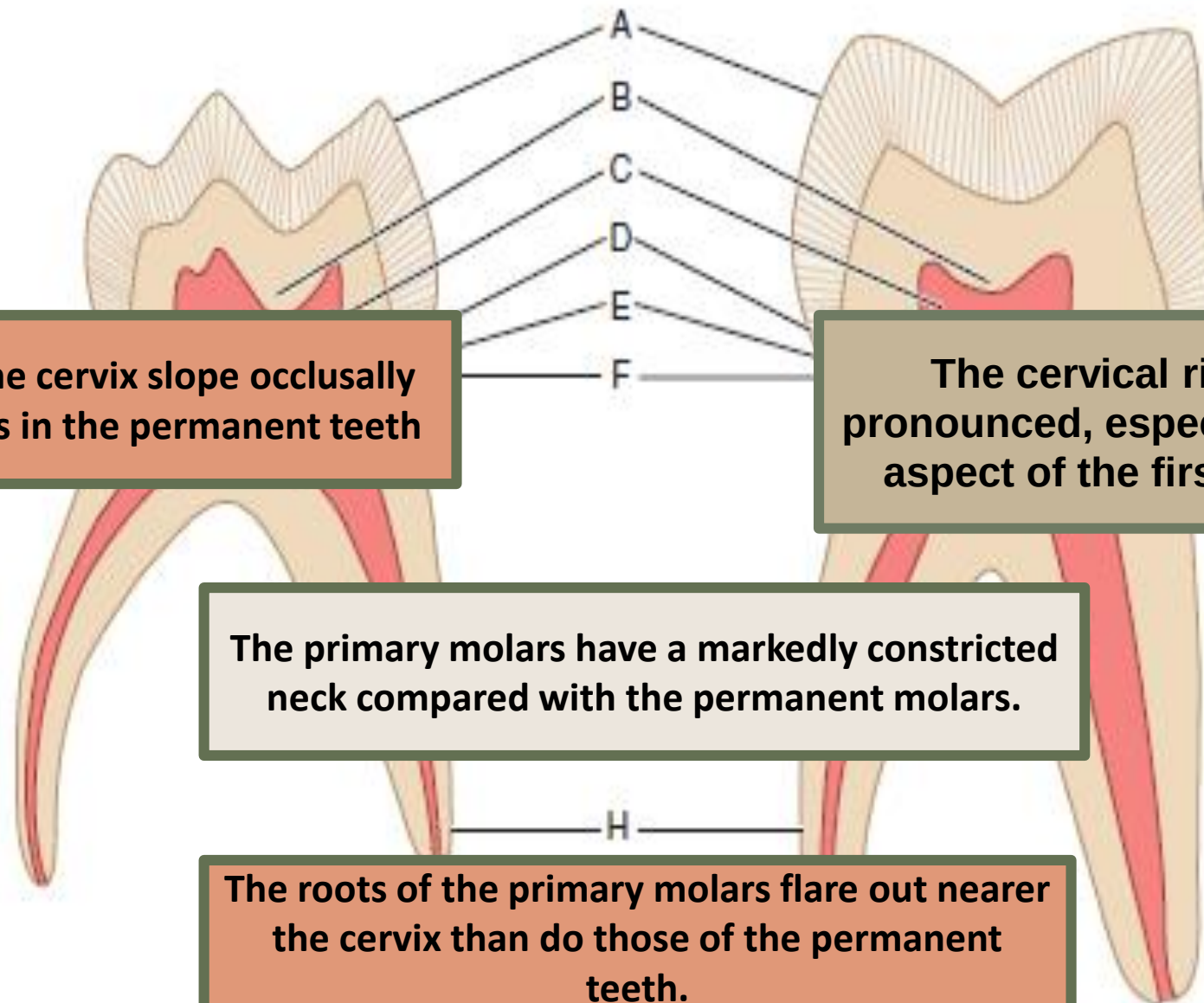
The enamel cap of primary molars is thinner and has a more consistent depth

A comparatively greater thickness of dentin is over the pulpal wall at the occlusal fossa of primary molars.

The roots of the primary teeth are longer and more slender in comparison with crown size than those of the permanent teeth.



The pulpal horns are higher in primary molars, especially the mesial horns, and pulp chambers are proportionately larger.



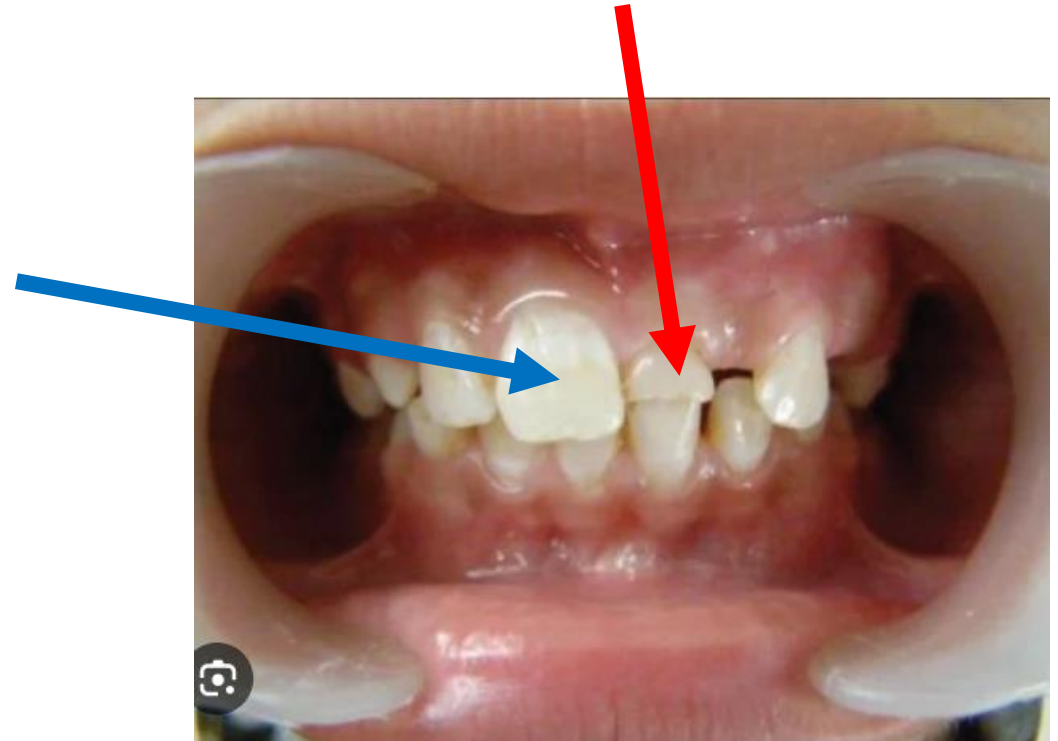
The enamel rods at the cervix slope occlusally instead of gingivally as in the permanent teeth

The cervical ridges are more pronounced, especially on the buccal aspect of the first primary molars

The primary molars have a markedly constricted neck compared with the permanent molars.

The roots of the primary molars flare out nearer the cervix than do those of the permanent teeth.

INDIVIDUAL TEETH COMPARATIVE DIFFERENCES BETWEEN PRIMARY AND PERMANENT TEETH



MAXILLARY CENTRAL INCISORS

MAXILLARY CENTRAL INCISORS – DIFFERENCES IN CHRONOLOGY

TOOTH	UNIVERSAL #	1 ST EVIDENCE OF CALCIFICATION	CROWN COMPLETION	ERUPTION	ROOT COMPLETION
Deciduous Maxillary Central Incisor	E, F	14 weeks IU	1 ½ months	10 months	1 ½ years

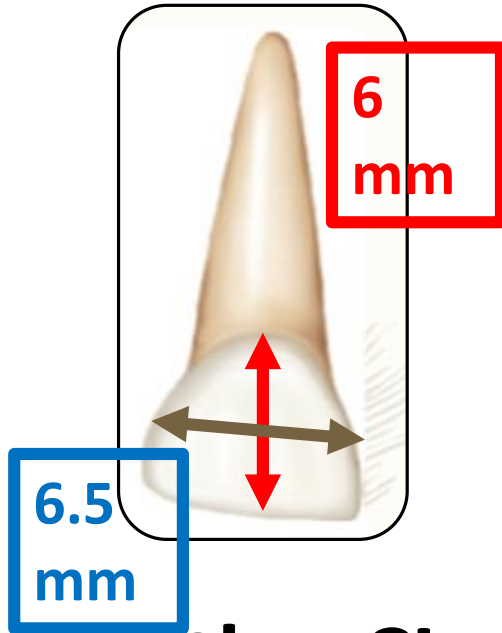
TOOTH	UNIVERSAL #	1 ST EVIDENCE OF CALCIFICATION	CROWN COMPLETION	ERUPTION	ROOT COMPLETION
Permanent Maxillary Central Incisor	8, 9	3-4 months	4-5 years	7-8 years	10 years

MAXILLARY CENTRAL INCISORS – DIFFERENCES IN DIMENSIONS

TOOTH	Deciduous Maxillary Central Incisor	Permanent Maxillary Central Incisor
Length of crown	6.0 mm	10.5 mm
Length of root	10.0 mm	13.0 mm
MD diameter of crown	6.5 mm	8.5 mm
MD diameter of crown at cervix	4.5 mm	7.0 mm
Labiolingual Diameter of crown	5.0 mm	7.0 mm
Labiolingual Diameter of crown at cervix	4.0 mm	6.0 mm

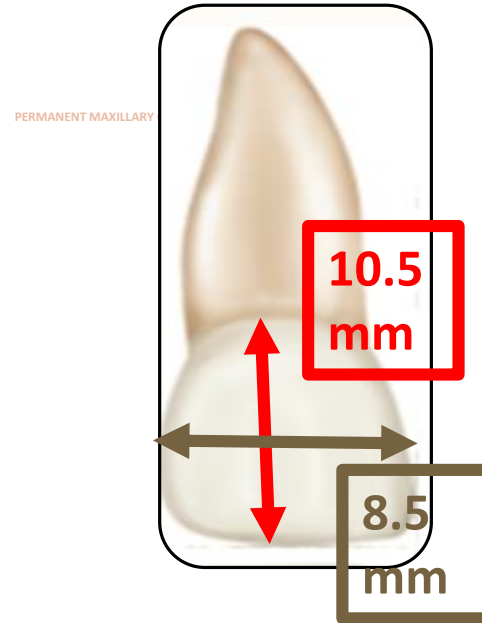
DIMENSIONS

PRIMARY MAXILLARY CENTRAL INCISOR



MD width > CI width

PERMANENT MAXILLARY CENTRAL INCISOR



CI width > MD width

LINGUAL ANATOMY

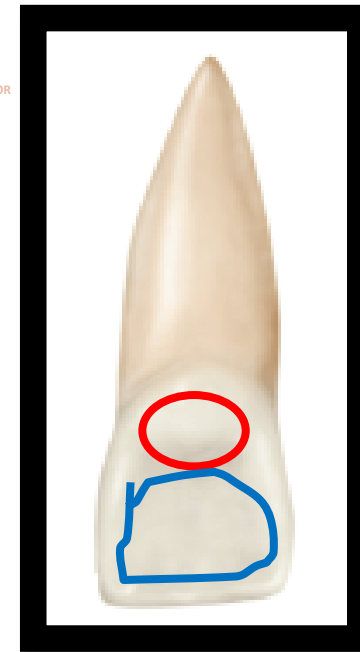
PRIMARY MAXILLARY CENTRAL INCISOR



**WELL DEVELOPED CINGULUM (EXTENDING
TOWARDS INCISAL RIDGE PARTIALLY
DIVIDING LINGUAL FOSSAE INTO M &
D FOSSA**

PERMANENT MAXILLARY CENTRAL INCISOR

PERMANENT MAXILLARY CENTRAL INCISOR



CINGULUM SMOOTH, SINGLE LINGUAL FOSSA

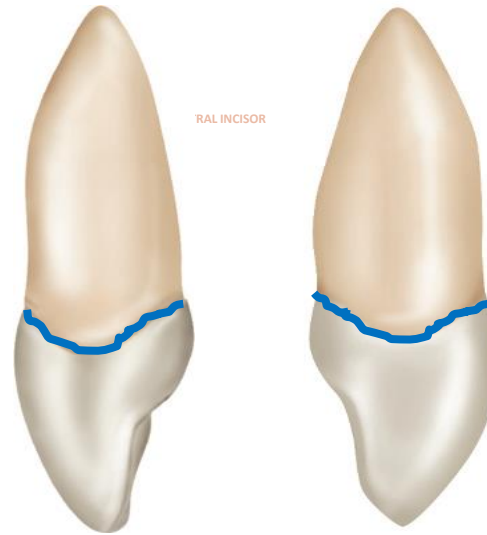
M & D ASPECTS

PRIMARY MAXILLARY CENTRAL INCISOR



**BOTH M & D, CERVICAL
LINE CURVATURE < PERMANENT
CERVICAL LINE CURVATURE**

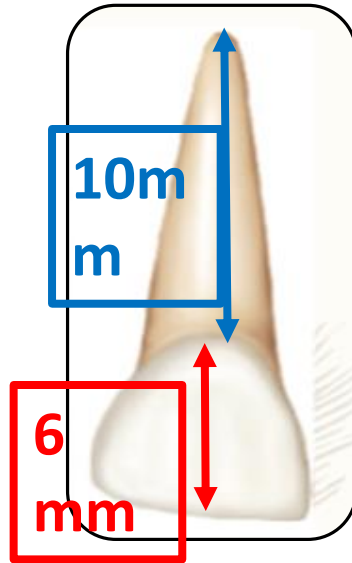
PERMANENT MAXILLARY CENTRAL INCISOR



**M CERVICAL LINE
CURVATURE
GREATEST OF ALL
TEETH IN MOUTH. D
CERVICAL LINE
CURVATURE < M, BUT
> PRIMARY
CURVATURES**

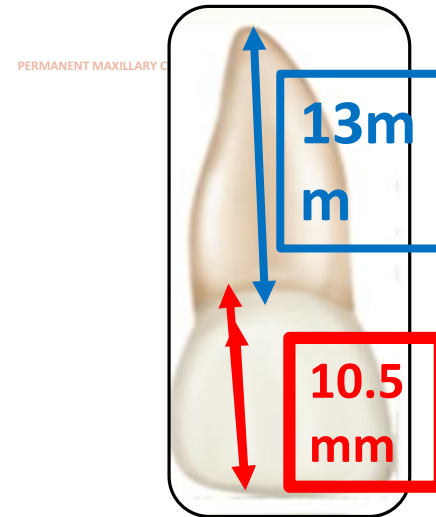
ROOT

PRIMARY MAXILLARY CENTRAL INCISOR



CONE-SHAPED WITH EVEN, TAPERED SIDES.
ROOT LENGTH COMPARISON WITH THE CROWN
LENGTH > ROOT LENGTH COMPARISON WITH
THE CROWN LENGTH OF PERMANENT

PERMANENT MAXILLARY CENTRAL INCISOR



CONE-SHAPED WITH EVEN M & D SIDE & BLUNT
APEX. ROOT IS 2-3 MM LONGER THAN CROWN



B



MAXILLARY LATERAL INCISORS



MAXILLARY LATERAL INCISORS – DIFFERENCES IN CHRONOLOGY

TOOTH	UNIVERSAL #	1 ST EVIDENCE OF CALCIFICATION	CROWN COMPLETION	ERUPTION	ROOT COMPLETION
Deciduous Maxillary Lateral Incisor	D, G	16 weeks IU	2 ½ months	11 months	2 years

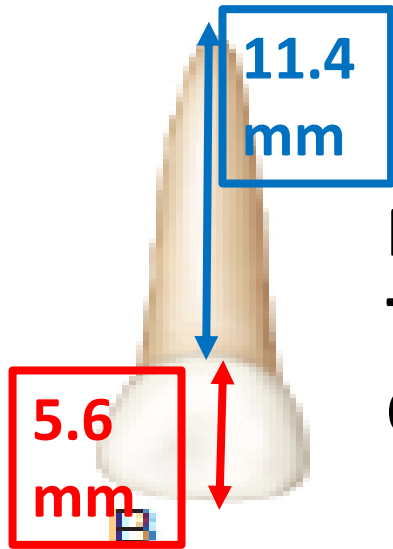
TOOTH	UNIVERSAL #	1 ST EVIDENCE OF CALCIFICATION	CROWN COMPLETION	ERUPTION	ROOT COMPLETION
Permanent Maxillary Central Incisor	7, 9	10-12 months	4-5 years	8-9 years	11 years

MAXILLARY LATERAL INCISORS – DIFFERENCES IN DIMENSIONS

TOOTH	Deciduous Maxillary Lateral Incisor	Permanent Maxillary Lateral Incisor
Length of crown	5.6 mm	9.0 mm
Length of root	11.4 mm	13.0 mm
MD diameter of crown	5.1 mm	6.5 mm
MD diameter of crown at cervix	3.7 mm	5.0 mm
Labiolingual Diameter of crown	4.0 mm	6.0 mm
Labiolingual Diameter of crown at cervix	3.7 mm	5.0 mm

ALL DIMENSIONS

PRIMARY MAXILLARY LATERAL INCISOR

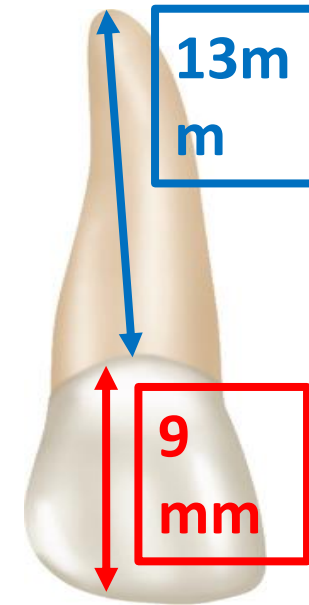


ROOT MUCH LONGER
THAN CROWN AS
COMPARED TO CI

LI CROWN DIMENSIONS > CI
CROWN DIMENSIONS, EXCEPT LL
(EQUAL IN BOTH)

PERMANENT MAXILLARY LATERAL INCISOR

PERMANENT MAXILLARY LATERAL INCISOR



ROOT IS
SLIGHTLY
LONGER
THAN
CROWN

LI CROWN DIMENSIONS > CI
CROWN DIMENSIONS ALL, EXCEPT
~~LL (EQUAL IN BOTH)~~

ALL DIMENSIONS

PRIMARY MAXILLARY LATERAL INCISOR



**VARIATIONS
UNCOMMON.**

PERMANENT MAXILLARY LATERAL INCISOR



MOST COMMONLY VARIABLE, AFTER 3RD MOLARS. PEG-SHAPED LATERALS, MISSING TOOTH, PRESENCE OF PALATOGINGIVAL GROOVE, POINTED TUBERCLE BEING PART OF CINGULUM, TWISTED ROOTS & DISTORTED CROWNS



MAXILLARY CANINE

MAXILLARY CANINES– DIFFERENCES IN CHRONOLOGY

Tooth	Universal #	1 st evidence of calcification	Crown completion	Eruption	Root completion
Deciduous Maxillary Canine	C, H	17 weeks IU	9 months	19 months	3 ¼ years

TOOTH	UNIVERSAL #	1 ST EVIDENCE OF CALCIFICATION	CROWN COMPLETION	ERUPTION	ROOT COMPLETION
Permanent Maxillary Canine	6, 11	4-5 months	6-7 years	11-12 years	13-15 years

MAXILLARY CANINE– DIFFERENCES IN DIMENSIONS

TOOTH	Deciduous Maxillary Canine	Permanent Maxillary Canine
Length of crown	6.5 mm	10.0 mm
Length of root	13.5 mm	17.0 mm
MD diameter of crown	7.0 mm	7.5 mm
MD diameter of crown at cervix	5.1 mm	5.5 mm
Labiolingual Diameter of crown	7.0 mm	8.0 mm
Labiolingual Diameter of crown at cervix	5.5 mm	7.0 mm

CUSP FEATURES & MESIAL CREST OF CONTOUR

PRIMARY MAXILLARY CANINE



**M CREST OF
CONTOUR MORE
CERVICAL THAN
PERMANENT.**

CUSP SHARPER & LONGER

PERMANENT MAXILLARY CANINE



**M CREST OF
CONTOUR
JUNCTION OF
MIDDLE & INCISAL
3RD.**

**CUSP SHORTER & MORE
BLUNT**

CONTACT AREAS

PRIMARY MAXILLARY CANINE



**LINE CONNECTING
CONTACT AREAS
BISECTS LINE FROM
CERVICAL LINE TO
CUSP TIP**

PERMANENT MAXILLARY CANINE

D-Mid 3rd

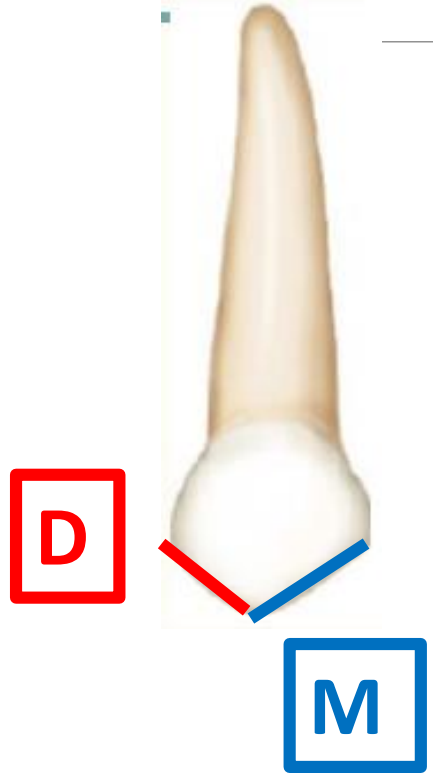


**M- junc
of middle
& incisal 3rd**

**M CROWN ½
CONTACTS WITH LI,
D ½ CONTACTS
WITH 1ST PM SO
CONTACT AREAS
DIFFERENT LEVEL**

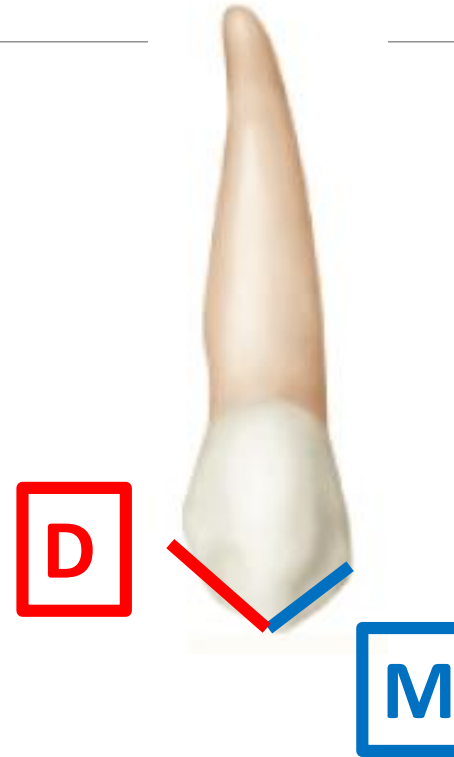
CUSPAL SLOPES

PRIMARY MAXILLARY CANINE



**M SLOPE > D
SLOPE, CUSP
TIP TOWARDS
DISTAL**

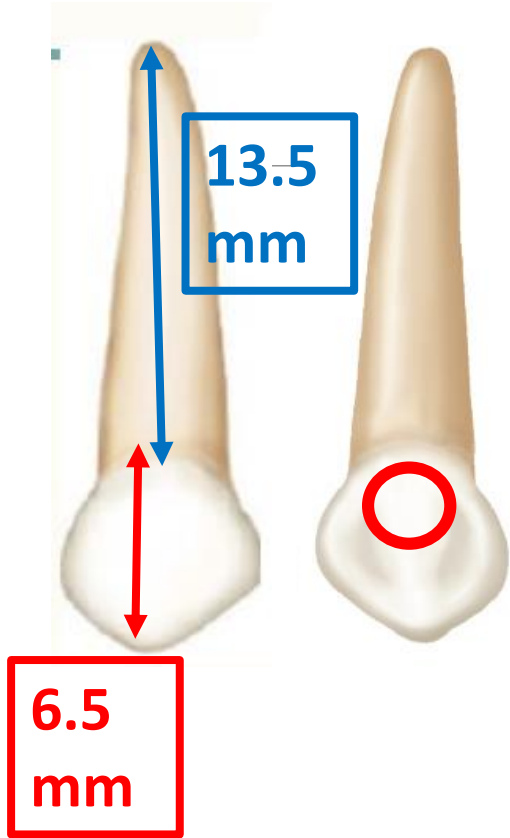
PERMANENT MAXILLARY CANINE



**M SLOPE < D SLOPE,
CUSP TIP TOWARDS
MESIAL**

LINGUAL ANATOMY & ROOT

PRIMARY MAXILLARY CANINE



**MORE
PRONOUNCED &
LESS ROUNDED
CINGULUM,
CERVICAL 3RD,
CONTACT AREAS,
LABIAL RIDGE &
CERVICAL 3RD.**

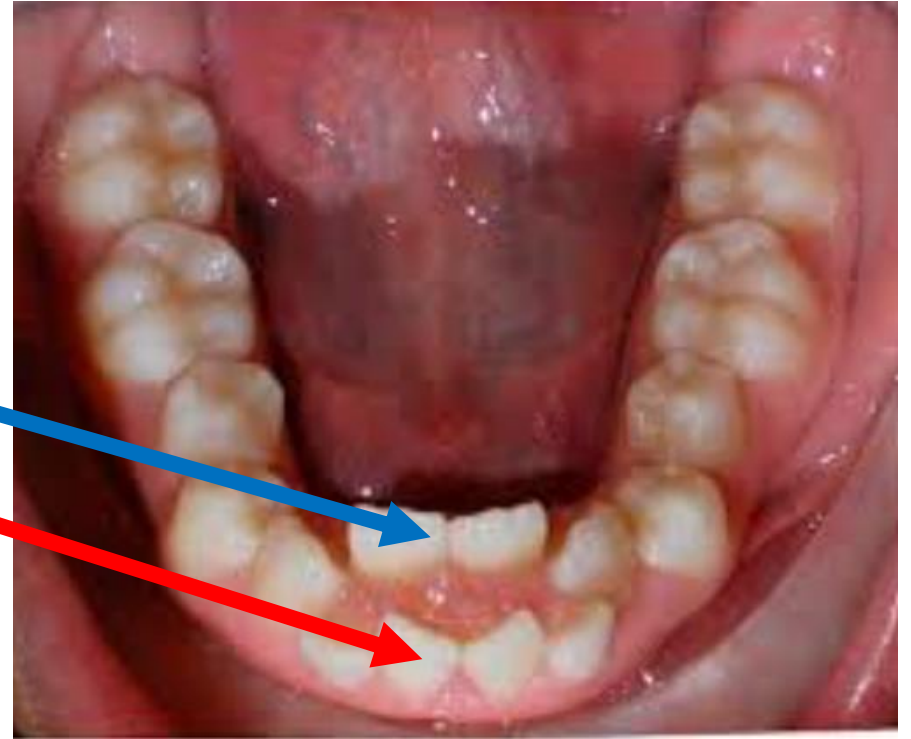
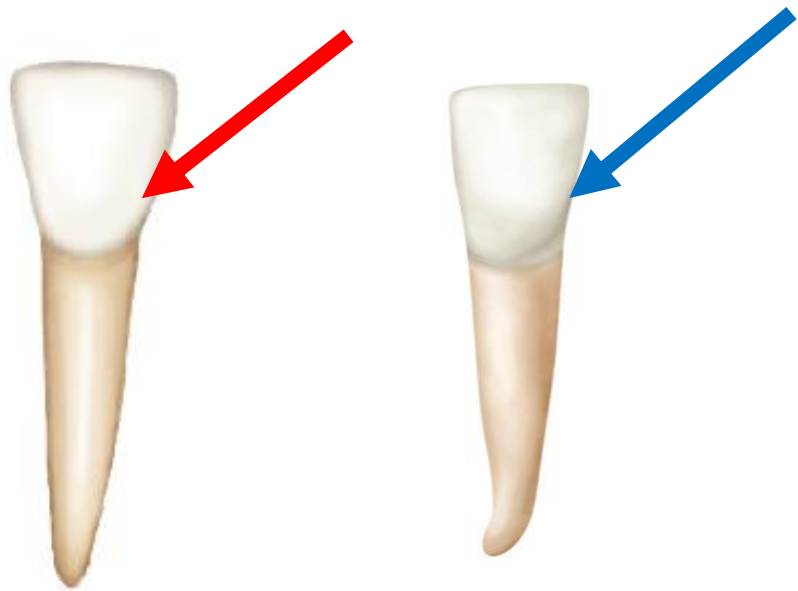
**ROOT LONG, SLENDER, TAPERING,
MORE THAN 2X CROWN LENGTH**

PERMANENT MAXILLARY CANINE



**LARGE &
POINTED
CINGULUM,
JUST LIKE A
CUSP.**

**LONGEST ROOT OF ALL TEETH. LL
THICK ROOT WITH M & D DEV.
DEPRESSIONS.**



MANDIBULAR CENTRAL INCISORS

MANDIBULAR CENTRAL INCISORS – DIFFERENCES IN CHRONOLOGY

TOOTH	UNIVERSAL #	1 ST EVIDENCE OF CALCIFICATION	CROWN COMPLETION	ERUPTION	ROOT COMPLETION
Deciduous Mandibular CI	P, O	14 weeks IU	2 ½ months	8 months	1 ½ years

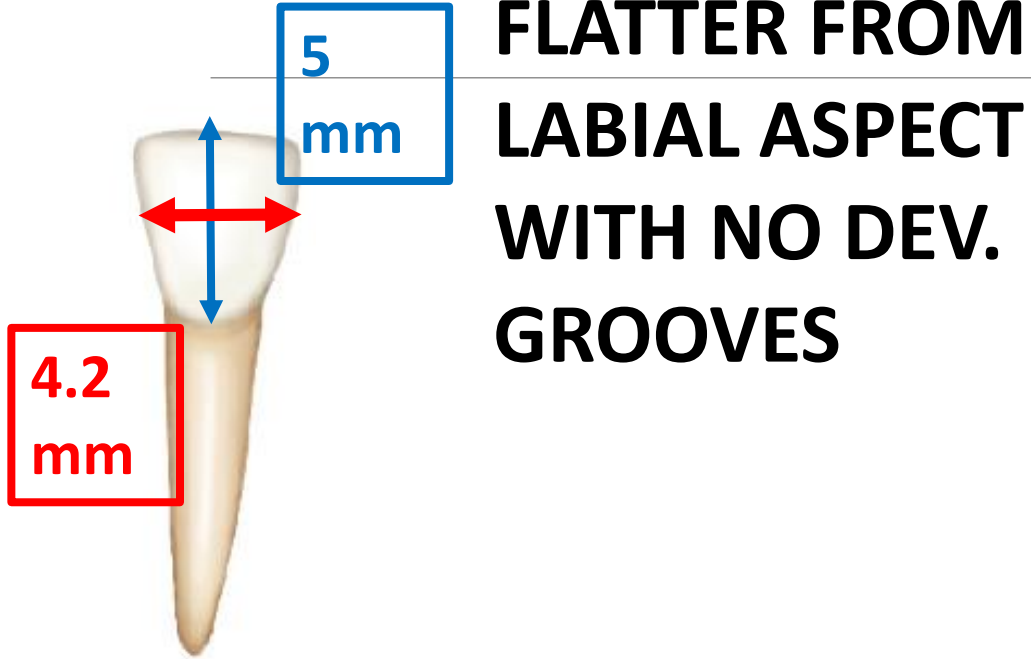
TOOTH	UNIVERSAL #	1 ST EVIDENCE OF CALCIFICATION	CROWN COMPLETION	ERUPTION	ROOT COMPLETION
Permanent Mandibular Central Incisor	24, 25	3-4 months	4-5 years	6-7 years	9 years

MANDIBULAR CENTRAL INCISORS – DIFFERENCES IN DIMENSIONS

TOOTH	Deciduous Mandibular Central Incisor	Permanent Mandibular Central Incisor
Length of crown	5.0 mm	9.5 mm
Length of root	9.0 mm	12.5 mm
MD diameter of crown	4.2 mm	5.0 mm
MD diameter of crown at cervix	3.0 mm	3.5 mm
Labiolingual Diameter of crown	4.0 mm	6.0 mm
Labiolingual Diameter of crown at cervix	3.5 mm	5.3 mm

CROWN FEATURES & PROPORTIONS

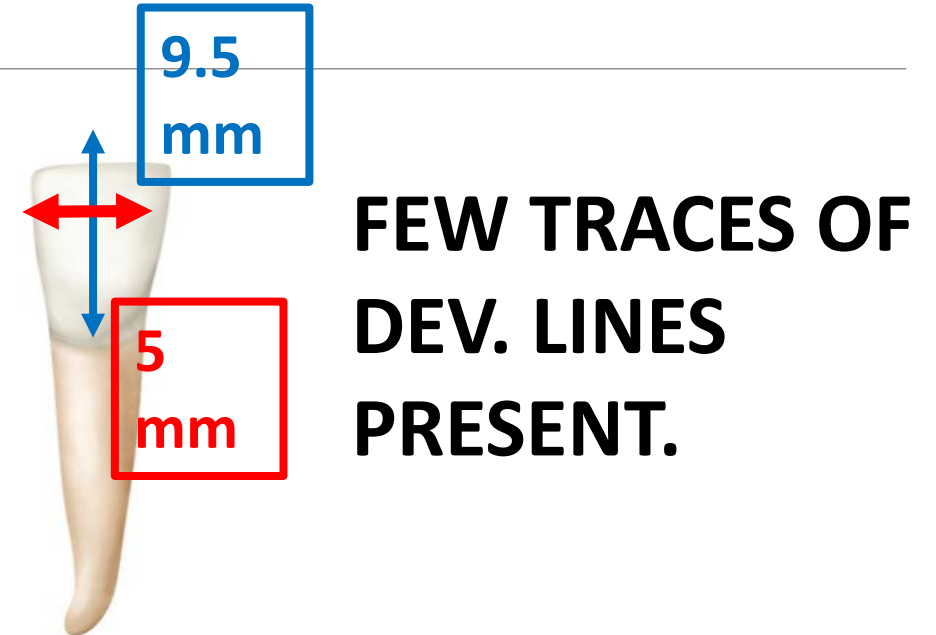
PRIMARY MANDIBULAR CENTRAL INCISOR



**FLATTER FROM
LABIAL ASPECT
WITH NO DEV.
GROOVES**

**CROWN WIDTH PROPORTION > LENGTH (ALTHOUGH
MD MEASUREMENT IS LESS, IT DOES NOT HAVE MUCH
DIFFERENCE AS COMPARED TO LENGTH) AS COMPARED
TO PERMANENT SUCCESSOR ONLY**

PERMANENT MANDIBULAR CENTRAL INCISOR

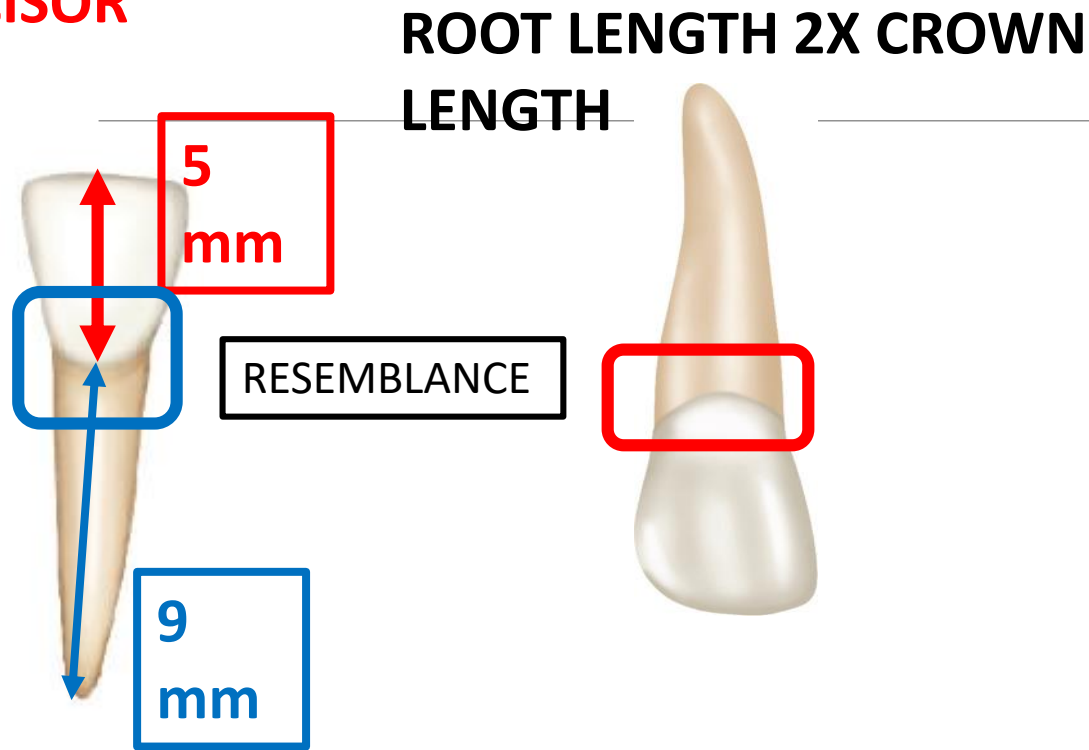


**FEW TRACES OF
DEV. LINES
PRESENT.**

CERVICO-INCISAL LENGTH ↑ > WIDTH

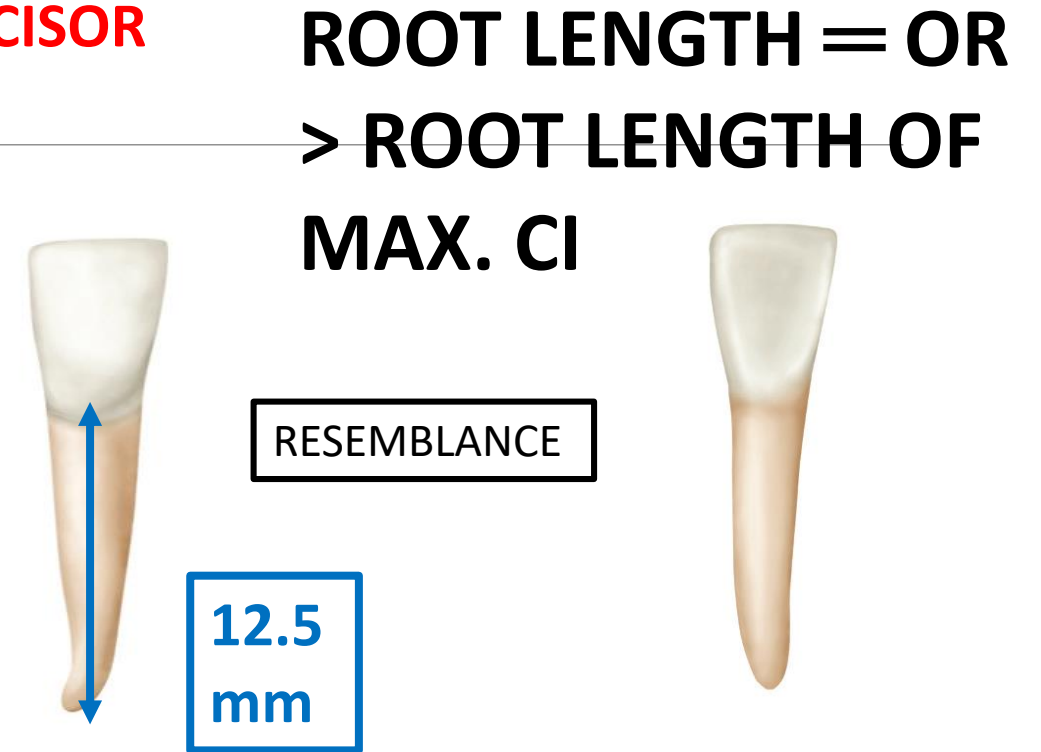
ROOT FEATURES & RESEMBLANCE

PRIMARY MANDIBULAR CENTRAL INCISOR

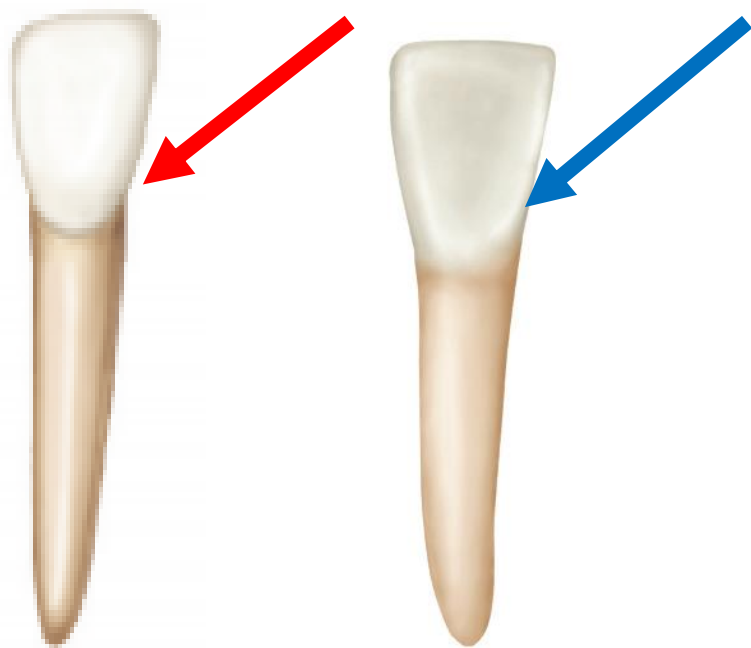


IT RESEMBLES PER. MAX. LI DUE TO HEAVY LOOK AT ROOT TRUNK (PER. LI ALSO LOOKS BULKY CERVICALLY WITH A ROUNDED CROWN MEETING ROOT BULK AS WITH PRI. MAND. CI'S ROOT WHICH IS BULKY & 2X CROWN LENGTH)

PERMANENT MANDIBULAR CENTRAL INCISOR



RESEMBLES PERMANENT MANDIBULAR LATERAL INCISOR.



MANDIBULAR LATERAL INCISORS

MANDIBULAR LATERAL INCISORS – DIFFERENCES IN CHRONOLOGY

TOOTH	UNIVERSAL #	1 ST EVIDENCE OF CALCIFICATION	CROWN COMPLETION	ERUPTION	ROOT COMPLETION
Deciduous Mandibular LI	Q, N	16 weeks IU	3 months	13 months	1 ½ years

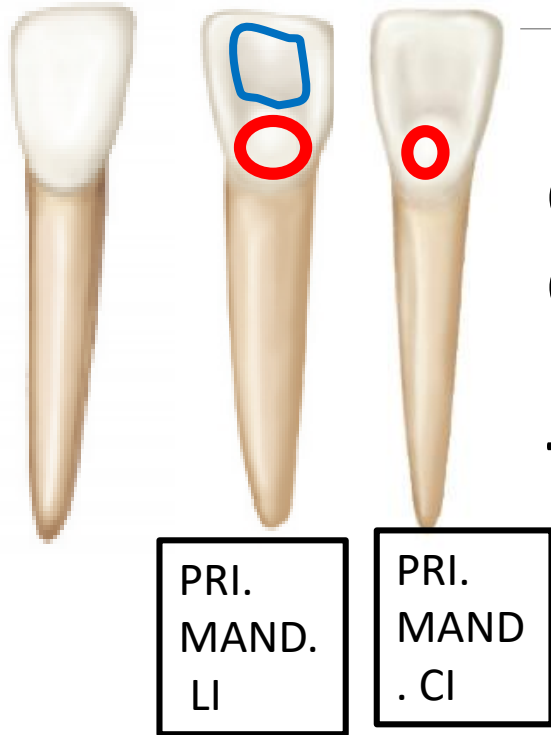
TOOTH	UNIVERSAL #	1 ST EVIDENCE OF CALCIFICATION	CROWN COMPLETION	ERUPTION	ROOT COMPLETION
Permanent Mandibular Lateral Incisor	23, 26	3-4 months	4-5 years	7-8 years	10 years

MANDIBULAR LATERAL INCISORS – DIFFERENCES IN DIMENSIONS

TOOTH	Deciduous Mandibular Lateral Incisor	Permanent Mandibular Lateral Incisor
Length of crown	5.6 mm	9.5 mm
Length of root	11.4 mm	14.0 mm
MD diameter of crown	5.1 mm	5.5 mm
MD diameter of crown at cervix	3.7 mm	4.0 mm
Labiolingual Diameter of crown	4.0 mm	6.5 mm
Labiolingual Diameter of crown at cervix	3.7 mm	5.8 mm

LINGUAL FEATURES & ROOT

PRIMARY MANDIBULAR LATERAL INCISOR



MORE PROMINENT CINGULUM & MORE CONCAVE LINGUAL FOSSA AS COMPARED TO MAND. CI

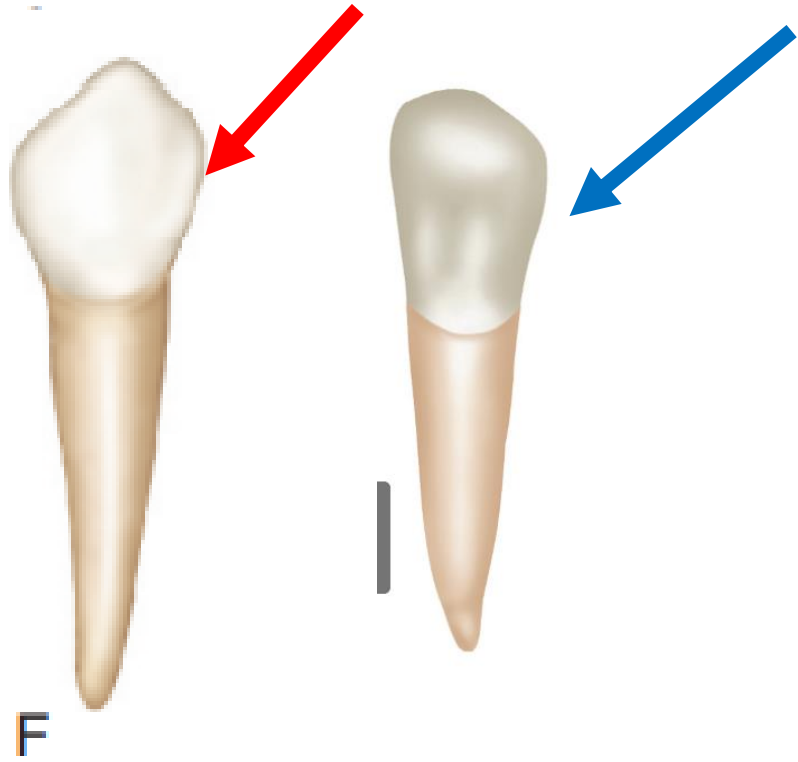
ROOT DOES NOT SHOW A DIVISION.

PERMANENT MANDIBULAR LATERAL INCISOR



CERVICO-INCISAL GROOVE MIGHT BE PRESENT IN MONGOLOID GROUPS LINGUALLY

LABIAL & LINGUAL ROOT DIVISION IN CERVICAL 3RD. 2 CANALS IN A SINGLE ROOT MORE COMMON.



MANDIBULAR CANINE

MANDIBULAR CANINES– DIFFERENCES IN CHRONOLOGY

Tooth	Universal #	1 st evidence of calcification	Crown completion	Eruption	Root completion
Deciduous Mandibular Canine	R, M	17 weeks IU	9 months	20 months	3 ¼ years

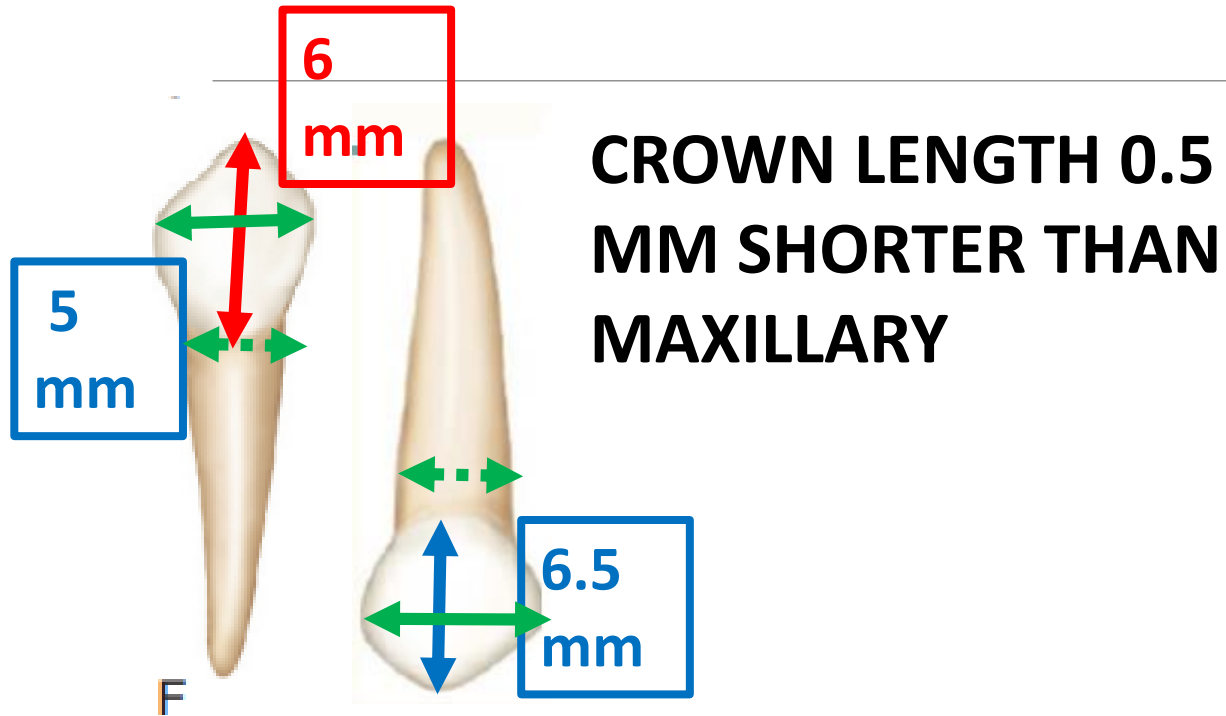
TOOTH	UNIVERSAL #	1 ST EVIDENCE OF CALCIFICATION	CROWN COMPLETION	ERUPTION	ROOT COMPLETION
Permanent Mandibular Canine	22, 27	4-5 months	6-7 years	9-10 years	12-14 years

MANDIBULAR CANINE– DIFFERENCES IN DIMENSIONS

TOOTH	Deciduous Mandibular Canine	Permanent Mandibular Canine
Length of crown	6.0 mm	11.0 mm
Length of root	11.5 mm	16.0 mm
MD diameter of crown	5.0 mm	7.0 mm
MD diameter of crown at cervix	3.7 mm	5.5 mm
Labiolingual Diameter of crown	4.8 mm	7.5 mm
Labiolingual Diameter of crown at cervix	4.0 mm	7.0 mm

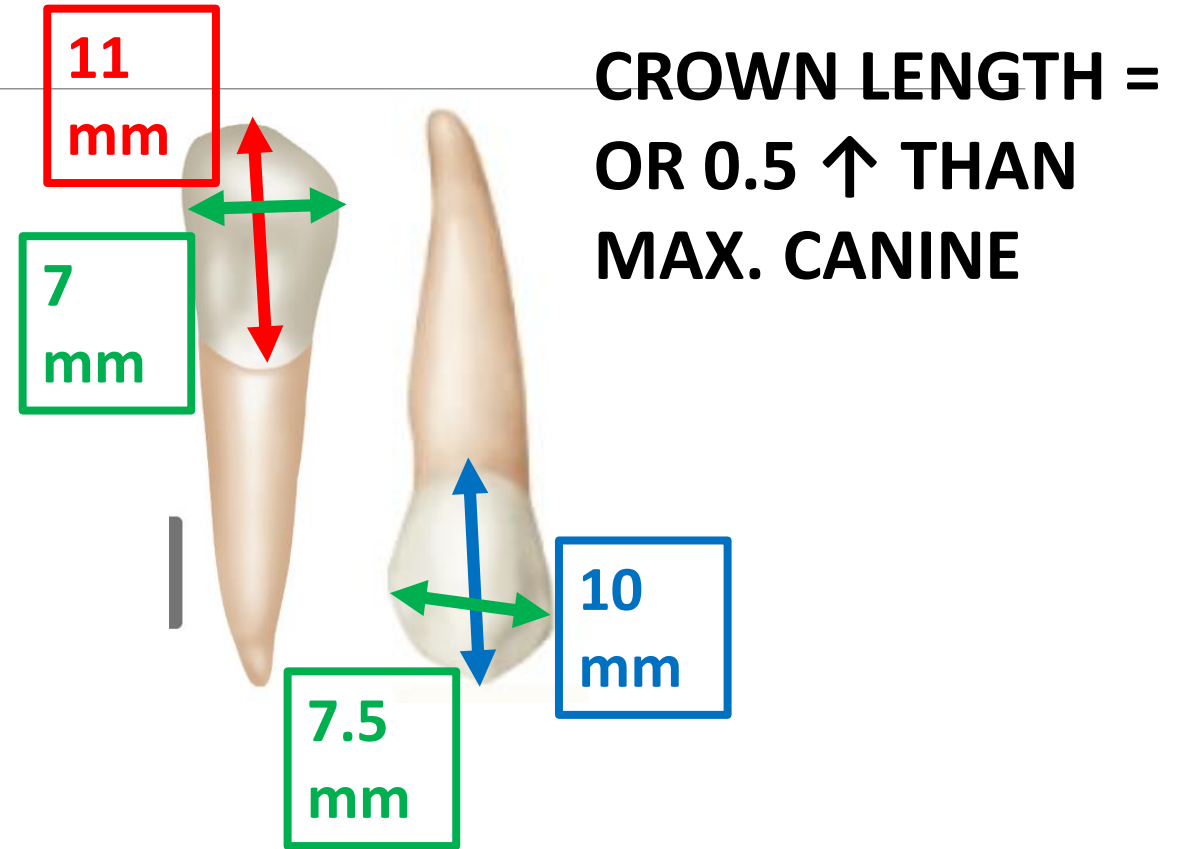
CROWN FEATURES & DIMENSIONS

PRIMARY MANDIBULAR CANINE



MD MEASUREMENT AT ROOT TRUNK >
MD MEASUREMENT AT CONTACT AREAS
AS COMPARED TO MAXILLARY CANINES
→ THICKER AT NECK OF TOOTH

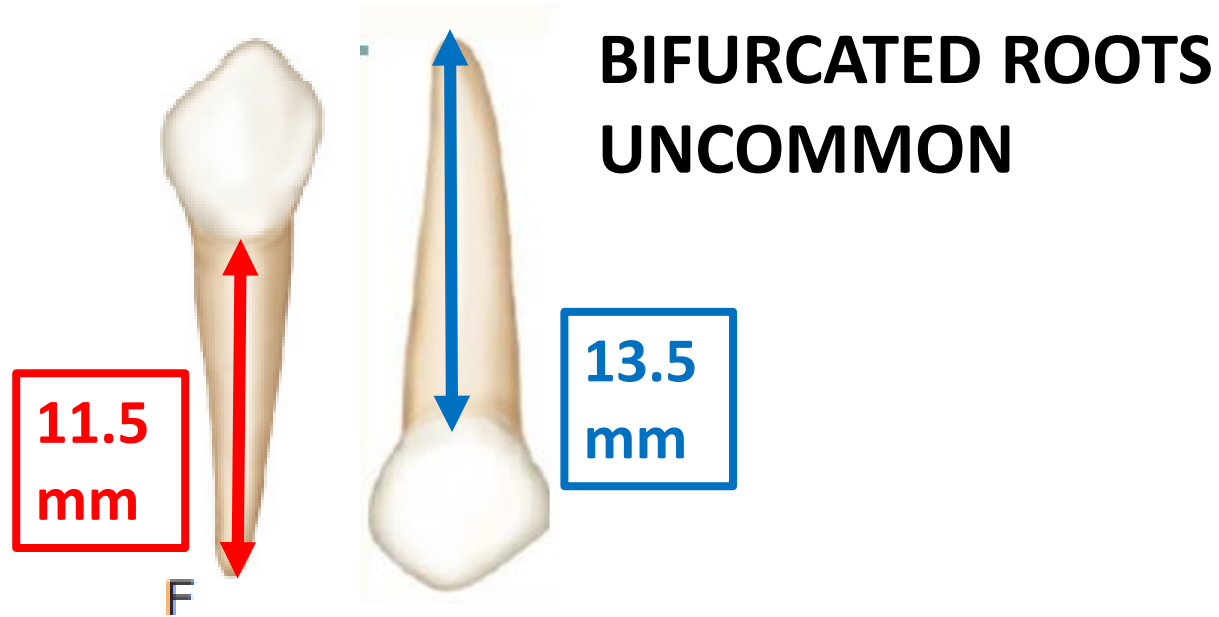
PERMANENT MANDIBULAR CANINE



MD WIDTH OF CROWN NARROWER THAN THAT OF MAXILLARY CANINE

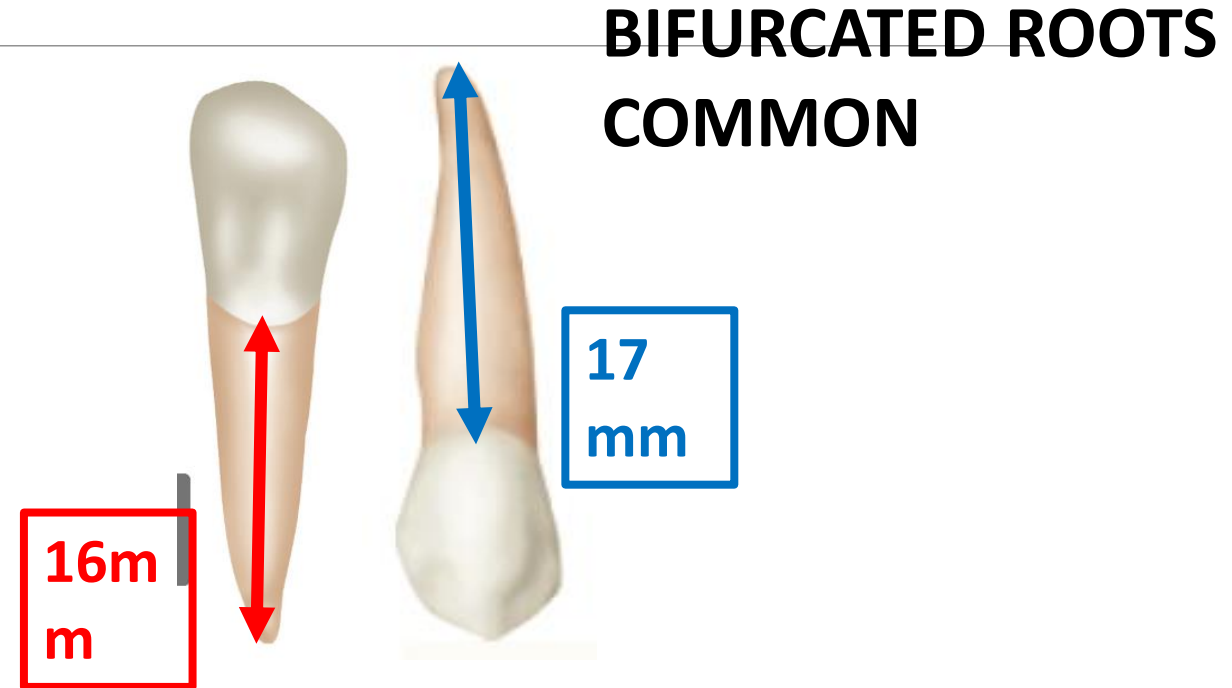
ROOT

PRIMARY MANDIBULAR CANINE



ROOT IS 2MM SHORTER THAN MAX. CANINE

PERMANENT MANDIBULAR CANINE



ROOT MIGHT BE AS LONG AS MAX CANINE BUT USUALLY IT IS SHORTER



**THANK
YOU**